Report on Train Fare Scraping Project

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1. The Chosen Topic

The project focuses on **scraping and analyzing train fare information** from publicly available websites. The goal is to extract train numbers, names, origins, destinations, and fare details for individual passengers based on different categories (e.g., sleeper, AC, general class). This data provides an insightful look into the cost structure of train travel across various routes in India.

2. The Website(s) Scraped

Websites Used:

1. **Prokerala Train Information** (`https://www.prokerala.com/travel/indian-railway/trains/`):
    - This site lists train numbers, names, and schedules, making it a starting point for identifying the trains for which fare details are required.
2. **Erail Train Fare Details** (`https://erail.in/train-fare/`):
    - This site provides detailed fare breakdowns for individual passengers under different fare categories, including Sleeper Class, AC Chair Car, and General Class, based on the selected origin and destination stations.

3. Relevance of the Extracted Data to the Goal

Objective:

The aim was to compile a structured dataset that encapsulates:

- Train numbers and names.
- Origin and destination stations.
- Individual passenger fares across various ticket categories.

Relevance:

The extracted data holds significance in:

- **Facilitating travel planning**: Passengers can compare fare categories and costs before booking.
- **Transportation analysis**: Analysts can evaluate fare structures and understand pricing trends for railway services.
- **Data-driven insights**: The dataset can be leveraged for machine learning models, such as predicting fare changes or optimizing travel budgets.
- **Policy insights**: It can assist policymakers in reviewing fare distribution for economic classes.

Additional Notes

Process Overview:

1. **Data Collection**:
    - Train numbers were fetched from Prokerala.
    - Fare details for each train were obtained by querying Erail's fare database.
2. **Data Cleaning & Structuring**:
    - The extracted data was organized into rows and columns using Python and Pandas.
    - Unique headers, such as fare categories, were derived and incorporated into the dataset.
3. **Output**:
    - The final output is a structured CSV file (`scrap.csv`) that includes comprehensive train and fare information.

Challenges Encountered:

- Websites with inconsistent HTML structures.
- Missing or incomplete data for certain train numbers.
- The need for dynamic handling of fare categories due to their variability.

Deliverables

Final Dataset:

- Contains details on train numbers, names, origins, destinations, and fare breakdowns.
- Saved as `scrap.csv` for further analysis.

Potential Extensions:

- Real-time fare updates using APIs.
- Visualization of fare trends and distributions.
- Integration with a travel recommendation system.

Conclusion

This project successfully demonstrates the use of web scraping and data processing to create a valuable dataset for railway fare analysis. The insights derived from the data can be applied to various domains, including travel planning, economic analysis, and public policy formulation.